SGN – Assignment #1

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1 Periodic orbit

Exercise 1

Consider the 3D Earth–Moon Circular Restricted Three-Body Problem with $\mu = 0.012150$.

1) Find the x-coordinate of the Lagrange point L_1 in the rotating, adimensional reference frame with at least 10-digit accuracy.

Solutions to the 3D CRTBP satisfy the symmetry

$$\mathcal{S} : (x, y, z, \dot{x}, \dot{y}, \dot{z}, t) \rightarrow (x, -y, z, -\dot{x}, \dot{y}, -\dot{z}, -t).$$

Thus, a trajectory that crosses perpendicularly the $y = 0$ plane twice is a periodic orbit.

2) Given the initial guess $\mathbf{x}_0 = (x_0, y_0, z_0, v_{x0}, v_{y0}, v_{z0})$, with

$$\begin{aligned} x_0 &= 1.08892819445324 \\ y_0 &= 0 \\ z_0 &= 0.0591799623455459 \\ v_{x0} &= 0 \\ v_{y0} &= 0.257888699435051 \\ v_{z0} &= 0 \end{aligned}$$

Find the periodic halo orbit that passes through z_0; that is, develop the theoretical framework and implement a differential correction scheme that uses the STM either approximated through finite differences or achieved by integrating the variational equation.

The periodic orbits in the CRTBP exist in families. These can be computed by continuing the orbits along one coordinate, e.g., z_0. This is an iterative process in which one component of the state is varied, while the other components are taken from the solution of the previous iteration.

3) By gradually decreasing z_0 and using numerical continuation, compute the families of halo orbits until $z_0 = 0.034$.

(8 points)

Part 1

For the calculation of Lagrange point L1, were used the partial derivatives of scalar potential function $\mathbf{U}$, taken with x,y,z.

$$U = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{1}$$

where r_1 and r_2 are the norm of the position vectors of Earth and Moon, and are given by:

$$r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2} \tag{2}$$

$$r_2 = \sqrt{(x+\mu-1)^2 + y^2 + z^2} \tag{3}$$

It is assumed that the three partial derivatives shall be 0, and from this is observed that $z = 0$, as long as $r_1 = r_2$. Furthermore, knowing that L_1, L_2 and L_3, have $y = 0$, it was calculated substituting the above results, into the initial equations (Eq. (1) and Eqs. (2), (3)). The obtained equation is given by Eq. (4).

$$\frac{\partial U}{\partial x}(x,y,z) = \frac{\partial U}{\partial x}(x,0,0) = x - (1-\mu)\frac{x+\mu}{|x+\mu|^3} - \mu\frac{x+\mu-1}{|x+\mu-1|^3} \tag{4}$$

The value of L_1 was obtained by using *fsolve*, with initial point 0, while for achieving a 10-digit accuracy, the *fsolve*'s tolerances were set to 10^{-10}.

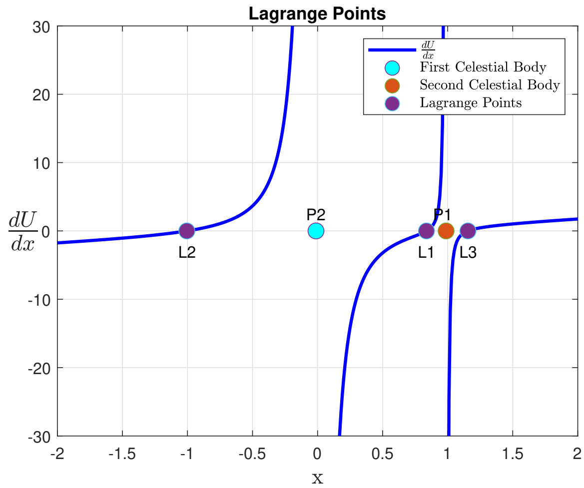


Figure 1: Locations of Lagrange points with respect to Partial derivative of Potential Function

Part 2

To determine the initial periodic halo orbit passing through z_0, a variational approach was deemed more suitable due to its higher precision within Newton's Method for differential correction. Given that differential correction is an iterative process, employing the variational approach ensures that the propagated error remains minimal. The dynamics of the body is the one of Circular Restricted 3 Body Problem. The State Transition Matrix was obtained by integrating, its derivative obtained from Eq. (5), where $\frac{\partial f}{\partial x}$ is the Jacobian of the flow ϕ, and can be noted also as $A(t)$.

$$\dot{\Phi} = \frac{\partial f}{\partial x}\Phi(t_0, t) \tag{5}$$

with

$$\Phi_0 = I_{6x6} \tag{6}$$

In order for the code to run smoother, a *syms* function was created in the beginning which initialized all the unknowns in Eq. (7), while the known values were taken at each instance inside the ode integrator.

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = x - \dfrac{1-\mu}{r_1^3}(\mu + x) + \dfrac{\mu}{r_2^3}(1-\mu-x) + 2v_y \\ \dot{v}_y = y - \dfrac{1-\mu}{r_1^3}y - \dfrac{\mu}{r_2^3}y - 2v_x \\ \dot{v}_z = \dfrac{-1+\mu}{r_1^3} - \dfrac{\mu}{r_2^3}z \end{cases} \tag{7}$$

The propagator that uses the Eq.(7),the Keplerian Dynamics, is accompanied by an event function that would stop the propagation when it intersects the x-axis. The initial state for the propagator is the one given in the problem's text.

Once the dynamics of the body was obtained and propagated, it was corrected through Newton's Method, as it was desired to have the velocity perpendicular to the x-axis, at $y = 0$.

$$\begin{pmatrix} \boldsymbol{\delta r_f} \\ \boldsymbol{\delta v_f} \end{pmatrix} = \begin{bmatrix} \Phi_{rr} & \Phi_{rv} \\ \Phi_{vr} & \Phi_{vv} \end{bmatrix} \begin{pmatrix} \boldsymbol{\delta r_0} \\ \boldsymbol{\delta v_0} \end{pmatrix} \tag{8}$$

The Newton's method was obtained from implementing Eq. (9), which would correct the orbit, starting from t_0 until it encounters the x-axis. As for obtaining the correction, just the values of v_y and x should be corrected, the system could be reduced to the one in (7). This is because the variations of v_{x0} and v_{z0} are equal to their respective flows. Thus the variation of the initial states is zero, except for those of v_{y0} and x_0. Therefore, the original system (8)(where δr represents the position variation and δv represents the velocity variation) is reduced to the following one, in which it is calculated the updated value for each step just for x_0 and v_{y0}. The updated values are calculated by using the corresponding terms of STM Φ, shown in Eq. (9).

$$\begin{pmatrix} \delta x_0 \\ \delta v_{y0} \end{pmatrix}_{k+1} = \begin{pmatrix} x_0 \\ v_{y0} \end{pmatrix}_k - \begin{pmatrix} \Phi_{41} & \Phi_{45} \\ \Phi_{61} & \Phi_{65} \end{pmatrix}^{-1} \begin{pmatrix} x_f \\ v_{yf} \end{pmatrix} \tag{9}$$

The tolerance for Newton's method is set at 10^{-10}, ensuring a high degree of accuracy in the solution. Additionally, a limit of 50 iterations is imposed on the *while-loop* to prevent excessive computational time and ensure the process terminates within a reasonable duration. This balance guarantees the precision desired in the results and efficiency in execution.

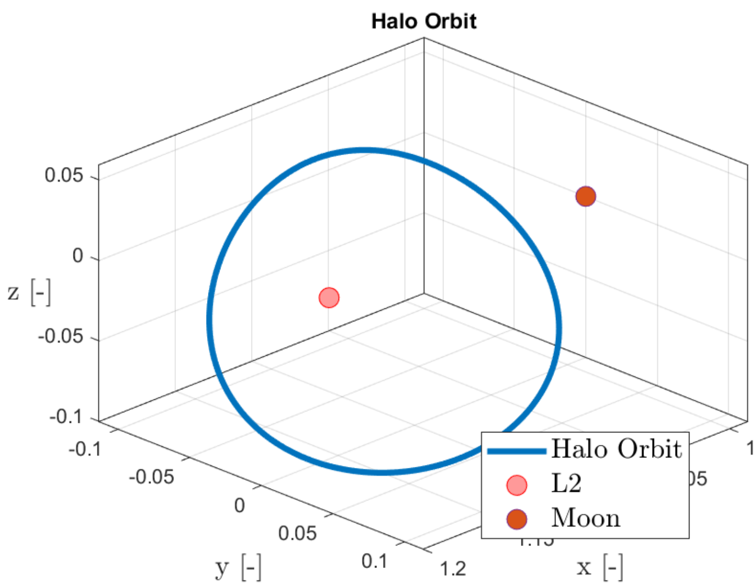


Figure 2: Halo orbit around the L_2

The initial state of the required orbit, after they have been corrected through Newton's Method, are reported in Table 1.

Part 3

For completing the third part of the exercise and ensure that the solution will converge, the method of numerical continuation was introduced in the process describe in **Part 2**. For this,

x_0	1.0902780546
y_0	0
z_0	0.0591799623
v_{x0}	0
v_{y0}	0.2603493850
v_{z0}	0

Table 1: Initial State of the plotted orbit

the z points were linearly spaced between the initial given value (z_0) and the requested value of 0.34. The process is carried over n times, n being the number of orbits desired, in this case 10 orbits were plotted. Thus the values of z_0 are taken from this vector, while the values of the other states are taken from the previous propagation, after they have been corrected through Newton's Method.

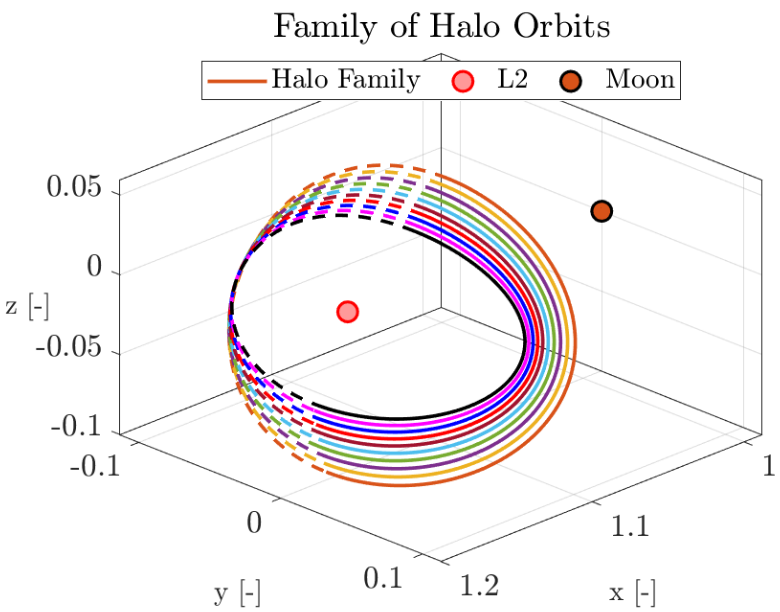


Figure 3: Family of Halo Orbits around L_2

Thus, the initial state of the final orbit plotted in the family of orbits 3, the required one that starts at $z_0 = 0.34$, has the initial coordinates, the following:

x_0	1.1116693684
y_0	0
z_0	0.034
v_{x0}	0
v_{y0}	0.2010186959
v_{z0}	0

Table 2: Initial values of the last Halo Orbit in the Family

2 Impulsive guidance

Exercise 2

The Aphophis close encounter with Earth will occur on April 2029. You shall design a planetary protection guidance solution aimed at reducing the risk of impact with the Earth.

The mission shall be performed with an impactor spacecraft, capable of imparting a $\Delta\mathbf{v} = 0.00005\,\mathbf{v}(t_{\text{imp}})$, where $\mathbf{v}$ is the spacecraft velocity and t_{imp} is the impact time. The spacecraft is equipped with a chemical propulsion system that can perform impulsive manoeuvres up to a total Δv of 5 km/s.

The objective of the mission is to maximize the distance from the Earth at the time of the closest approach. The launch shall be performed between 2024-10-01 (LWO, Launch Window Open) and 2025-02-01 (LWC, Launch Window Close), while the impact with Apophis shall occur between 2028-08-01 and 2029-02-28. An additional Deep-Space Maneuver (DSM) can be performed between LWO+6 and LWC+18 months.

1) Analyse the close encounter conditions reading the SPK kernel and plotting in the time window [2029-01-01; 2029-07-31] the following quantities:
   a) The distance between Apophis and the Sun, the Moon and the Earth respectively.
   b) The evolution of the angle Earth-Apophis-Sun
   c) The ground-track of Apophis for a time-window of 12 hours centered around the time of closest approach (TCA).
2) Formalize an unambiguous statement of the problem specifying the considered optimization variables, objective function, the linear and non-linear equality and inequality constraints, starting from the description provided above. Consider a multiple-shooting problem with $N = 3$ points (or equivalently 2 segments) from t_0 to t_{imp}.
3) Solve the problem with multiple shooting. Propagate the dynamics of the spacecraft considering only the gravitational attraction of the Sun; propagate the post-impact orbit of Apophis using a full n-body integrator. Use an event function to stop the integration at TCA to compute the objective function; read the position of the Earth at t_0 and that of Apophis at t_{imp} from the SPK kernels. Provide the optimization solution, that is, the optimized departure date, DSM execution epoch and the corresponding $\Delta\mathbf{v}$'s, the spacecraft impact epoch, and time and Distance of Closest Approach (DCA) in Earth radii. Suggestion: try different initial conditions.

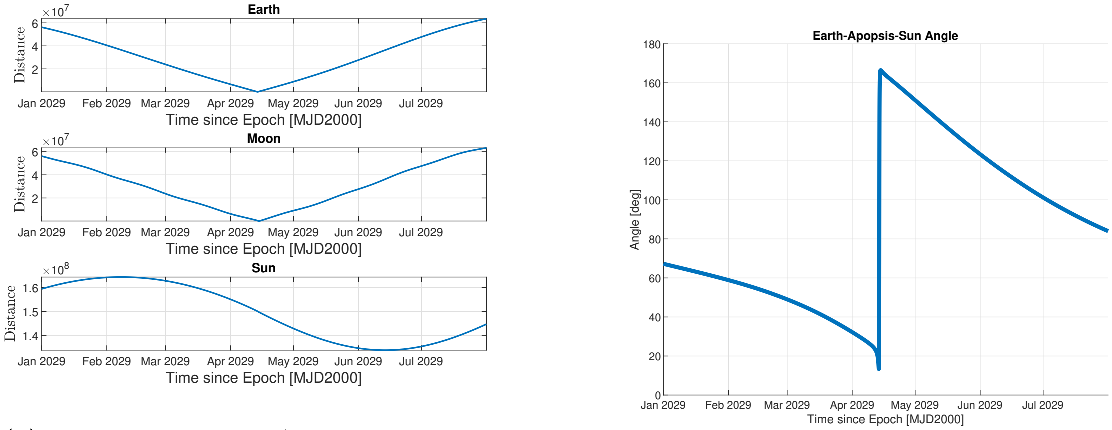
(11 points)

Part 1

The distances between Apophis and the three celestial bodies was calculated through the kernels provided, by using *cspice_spkpos*, and considering as reference frame the Earth rotating reference frame, for each body, and for the requested time window. The results can be seen in the Fig. 4a.

The Earth-Apophis-Sun angle was computed from the distances obtained previously and taking from kernels also the distance Sun-Earth. From these distances was obtained the angle at each time step. The angle evolution over time can be seen in Fig. 4b.

To plot the ground track, the latitude and longitude were computed using *cspice_reclat*. These coordinates were derived from the evolving position of Apophis relative to Earth, considering the minimum distance between the two objects to define the time window. The resulting ground track is shown in Fig. 5. The Apophis reaches the closest point to Earth on 2029 APR 13, at 21:44:58.



(a) Distances Between Apophis and Earth, Moon and Sun [km]

(b) Angle Earth-Apophis-Sun

Figure 4: Apophis in relation with other celestial bodies

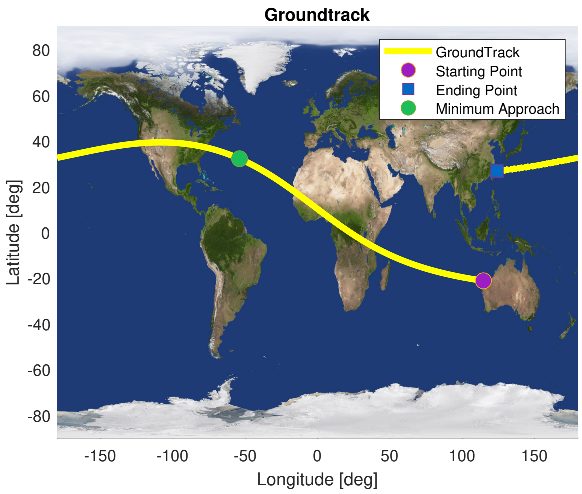


Figure 5: Apophis GroundTrack

Part 2

The optimization variables considered are:

- $\mathbf{x_1}$ - the initial states of the spacecraft at the launch
- $\mathbf{x_2}$ - the spacecraft's states at the beginning of the deep space maneuver
- $\mathbf{x_3}$ - the spacecraft's states at the impact with the Apophis
- $\mathbf{T}$ - composed of initial time of launch t_1, the time at which the deep space maneuver takes place t_2 and the time of impact with Apophis t_3

However, as the *Matlab* function, *fmincon*, accepts just one vector of initial guess, the optimization variables have been concatenated in a vector of size 21×1, containing all the variables, so the function a similar vector as output.

The objective function of the problem is trying to minimize the closest approach between Earth and Apophis, thus finding the optimal states at which each maneuver takes place and their epochs. Moreover, at least two steps should be considered, as it shall still respect the requirements of being a multiple-shooting method. The objective function provided in Eq. (10), is derived from the general impulsive guidance problem statement, which have been adapted to the present problem.

The equality constraints are those relating the states of the spacecraft, meaning that each shooting should start at the end of the previous one or at the position of the celestial body, as stated in Eq. (10). The inequality constraints are those relating to the time and as requested to the total Δv of the mission, which are shown as well in the Eq. (10)

$$min \boldsymbol{\Delta}\mathbf{r}(\mathbf{x}) \; s.t. \begin{cases} |\boldsymbol{\Delta}\mathbf{v_1}| + |\boldsymbol{\Delta}\mathbf{v_2}| \leq 5 \\ LWO \leq t_1 \\ LWC \geq t_1 \\ DSMWO \leq t_2 \\ DSMWC \geq t_2 \\ IWO \leq t_3 \\ IWC \geq t_3 \\ \mathbf{r_1} - \mathbf{r_{Earth}} = 0 \\ \boldsymbol{\varphi_1} - \mathbf{r_2} = 0 \\ \boldsymbol{\varphi_2} - \mathbf{x_3} = 0 \\ \mathbf{r_3} - \mathbf{r_{Apophis}} = 0 \end{cases} \tag{10}$$

In Eq. (10) LWO and LWC represent the time-window limits for launch, $DSMWO$ and $DSMWC$ the time-window limits for the deep space maneuver while IWO and IWC the time-window limits for impact. $\mathbf{r_{Earth}}$ and $\mathbf{r_{Apophis}}$ are the positions of the celestial bodies at launch, and at impact respectively, while $\mathbf{r_1}$ and $\mathbf{r_3}$ are the spacecraft's position at launch and at impact. $\mathbf{r_2}$ and $\mathbf{x_3}$ are the states of the spacecraft during the deep-space maneuver and at impact.

Let $\boldsymbol{\varphi}$ be the propagated state of an initial value, then $\boldsymbol{\varphi_1}(\mathbf{x_1}, t_0, t_{DSM})$ is the flow of the spacecraft from launching to deep space maneuver and $\boldsymbol{\varphi_2}(\mathbf{x_2}, t_{DSM}, t_{IMP})$ is the flow from the deep space maneuver to the impact with the asteroid.

In this case $\mathbf{r_{Earth}}$ and $\mathbf{r_{Apophis}}$, are the boundary condition of the problem as they are obtained from the kernels.

Part 3

For solving the problem the above procedure was implemented, considering the initial guess for the states as follows:

- $\mathbf{x_1}$ - The state during the launch, is the x_{Earth}, the state of the Earth at which $\boldsymbol{\Delta v}$ was added, in order to have a velocity greater than that of Earth's.

- $\mathbf{x_2}$ - Obtained from propagating the state at launch, at which was added a second $\boldsymbol{\Delta v}$, which would explain the change in velocity during the deep-space maneuver.

- $\mathbf{x_3}$ - The state at the impact, obtained from propagating $\mathbf{x_2}$.

The upper and lower limits considered for the *fmincon*, were taken as values large enough that would contain the final values obtained, thus in this case were considered to be equal to 10^{-21}. Regarding the initial times, guesses were taken proposing a value by choice, that would place the guesses in the time windows for each manoeuvre: $T_{guess} = TWO + \alpha(TWC - TWO)$, where TWO represents the opening date of the time window and TWC the closing date of the time window. This value was obtained after a number of trials of changing the multiplication factor, and in the end the value of $\alpha = \dfrac{1}{3}$ was the one leading to an optimal solution that would conclude in a Δv smaller than 5. The $\boldsymbol{\Delta v}$ that were added to the initial guesses, were calculated on a random bases, and they had to respect the general Δv constraints, imposed by the problem. The results in Table 3 shows the results obtained for the multiple-shooting problem, solving the NLP. Figure 6 shows the distance between Earth and Apophis taken from kernels (blue line) and the evolution of the distance between the two bodies in the interception case(orange line).

Launch	2024-NOV-11-20:53:15.300 UTC		
DSM	2026-JAN-11-01:48:56.773 UTC		
Impact	2028-NOV-13-21:18:59.464 UTC		
TCA	2029-APR-14-01:21:07.781 UTC		
$\Delta\mathbf{v}_L$ [km/s]	-1.45	2.33	-1.48
$\Delta\mathbf{v}_{DSM}$ [km/s]	-1.68	-0.03	-0.84
DCA [Re]	-15.71		

Table 3: Guidance solution for the impactor mission.

It must be noted that a better result could have been obtained regarding the minimum distance, but a much higher Δv would have been required, an thus this is the optimal solution. Another approach would be to optimize also the initial guesses, the α coefficients for the time guesses, and the Δv for the initial states guesses. This would lead to an improved solution, but would increases also the computational complexity.

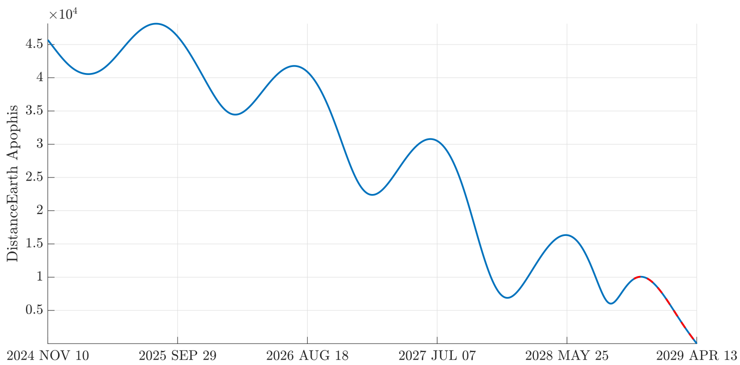


Figure 6: Earth-Apophis Distance in Earth Radii(MJD2000)

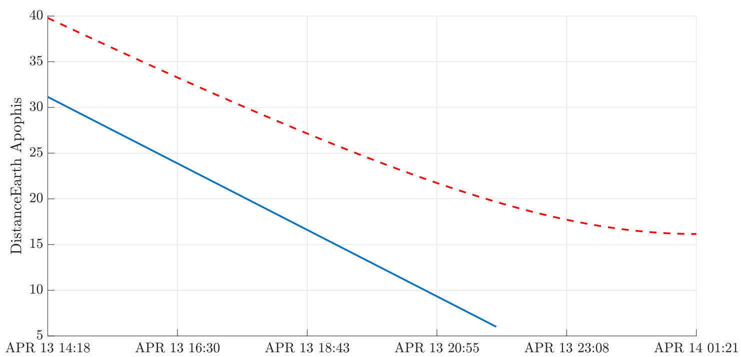


Figure 7: Earth-Apophis Distance in Earth Radii, at impact(MJD2000)

3 Continuous guidance

Exercise 3

A low-thrust option is being considered for an Earth-Venus transfer*. Provide a *time-optimal* solution under the following assumptions: the spacecraft moves in the heliocentric two-body problem, Venus instantaneous acceleration is determined only by the Sun gravitational attraction, the departure date is fixed, and the spacecraft initial and final states are coincident with those of the Earth and Venus, respectively.

1) Using the PMP, write down the spacecraft equations of motion, the costate dynamics, and the zero-finding problem for the unknowns $\{\boldsymbol{\lambda}_0, t_f\}$ with the appropriate transversality condition.

2) Adimensionalize the problem using as reference length $LU = 1$ AU† and reference mass $MU = m_0$, imposing that $\mu = 1$. Report all the adimensionalized parameters.

3) Solve the problem considering the following data:

   - Launch date: `2023-05-28-14:13:09.000 UTC`
   - Spacecraft mass: $m_0 = 1000$ kg
   - Electric propulsion properties: $T_{\max} = 800$ mN, $I_{sp} = 3120$ s

   To obtain an initial guess for the costate, generate random numbers such that $\lambda_{0,i} \in [-20; +20]$, while $t_f < 2\pi$. Report the obtained solution in terms of $\{\boldsymbol{\lambda}_0, t_f\}$ and the error with respect to the target. Assess your results exploiting the properties of the Hamiltonian in problems that are not time-dependent and time-optimal solutions.

4) Solve the problem for a lower thrust level $T_{\max} = [500]$ mN. Tip: exploit numerical continuation.

(11 points)

Part 1

By using PMP the *time-optimal* problem can be written in the following form as represented by Eq. (11), considering that the spacecraft is constrained by the position and velocity of the two planets, Earth at t_0 and Venus at t_f. The objective function J, is written in Lagrange form.

$$\min_{\mathbf{u}\,\in\,\overline{U}} J = \int_{t_f}^{t_0} 1\,dt \;\; s.t. \;\; \begin{cases} \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, t) \\ \mathbf{x}(t_0) = \mathbf{x_0} \\ \mathbf{r}(\mathbf{t_f}) = \mathbf{r_f} \\ \mathbf{v}(\mathbf{t_f}) = \mathbf{v_f} \\ \lambda_{\mathbf{m}}(\mathbf{t_f}) = 0 \end{cases} \tag{11}$$

where $\overline{U}$ is the set of admissible controls, and has the expression written in Eq. (12), where u is the throttle factor and $\hat{\alpha}$ is the direction unit vector.

$$\overline{U} = \{(u, \hat{\alpha}) \; s.t. \; u \in [0,1], ||\hat{\alpha}|| = 1\} \tag{12}$$

The equations of motion are given by the equation of motions of the two-body problem, in which thrust is generated by the engines, thus they involve also the evolution of mass in time,

*Read the necessary gravitational constants and planets positions from SPICE. Use the kernels provided on WeBeep for this assignment.

†Read the value from SPICE

as described in Eq. (13). These are obtained from the partial derivative of the Hamiltonian, H, with the costate vector λ , $\dot{\mathbf{x}} = \dfrac{\partial H}{\partial \lambda}$

$$\begin{cases} \dot{\mathbf{r}} = \mathbf{v} \\ \dot{\mathbf{v}} = -\dfrac{\mu}{r^3}\mathbf{r} + u^* \dfrac{T_{max}}{m}\hat{\alpha}^* \\ \dot{\mathbf{m}} = -u^* \dfrac{T_{max}}{I_{sp} g_0} \end{cases} \tag{13}$$

The costate dynamics is given by the partial derivative of the Hamiltonian with the state vector $\dot{\lambda} = -\dfrac{\partial H}{\partial \mathbf{x}}$. Moreover, in the systems (13) and (14) the throttle factor is taken as $u^* = 1$, as it was observed during the computations that the switching function ,S_t is always smaller than 1,while the direction unit factor is considered to be $\hat{\alpha}^* = -\dfrac{\lambda_v}{\lambda_v}$. *Pontryagin Maximum Principle* states that for an optimal solution to be found, Eq. (15) must be respected

$$\begin{cases} \dot{\lambda_{\mathbf{r}}} = -3\dfrac{\mu}{r^5}(\mathbf{r} \cdot \lambda_{\mathbf{v}}) + \dfrac{\mu}{r^3}\lambda_{\mathbf{v}} \\ \dot{\lambda_{\mathbf{v}}} = -\lambda_{\mathbf{r}} \\ \dot{\lambda_{\mathbf{m}}} = -u^* \dfrac{\lambda_{\mathbf{v}} T_{max}}{m^2} \end{cases} \tag{14}$$

$$\mathbf{u} = \arg \min_{\mathbf{u} \in \overline{U}} H(\mathbf{x}, \lambda, \mathbf{u}) \tag{15}$$

The Hamiltonian, H, on which are based the above equations has the expression in Eq. (16), while in the Eq. (17), it is shown the Transversality condition needed to impose another boundary condition for the values of λ

$$H = l + \lambda \mathbf{f} \tag{16}$$

$$H(t_f) - [\lambda_r(t_f)\dot{\Psi}_r(t_f) + \lambda_v(t_f)\dot{\Psi}_v(t_f)] = 0 \tag{17}$$

In Eq. (17), Ψ represents the state vectors of the Venus at interception time, with the subscripts indicating the position and the velocity vectors.

Part 2

In order to adimensionalize the problem, all the units provided must be converted with respect to the adminesionalize values. Moreover, as the reference length and μ have been converted, also the time should be adimensionalized, such that $TU = \sqrt{\dfrac{LU^3}{\mu}}$. The obtained adimenstionalized values are shown in Table 4

$\mathbf{r_0}$	−0.4009314	−0.9306291	+0.0000487
$\mathbf{v_0}$	+0.9296837	−0.3687609	0.0043055
m_0	1		
I_{sp}	0.00062118		
T_{max}	0.1349053		
g_0	1654.276		
GM	1		

Table 4: Adimensionalized quantities ($T_{\text{max}} = 800$ mN).

Part 3

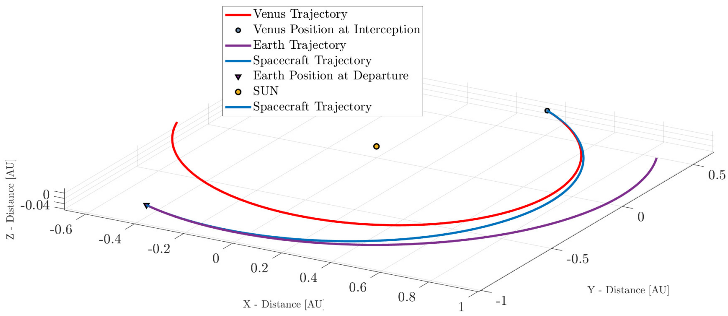


Figure 8: The 3 orbits for a $T_{max} = 800mN$(@SUN ECLIPJ2000)

In order to solve the problem, the equations described in **Part 1** have been used. However, as the initial costate were guesses, the correct values of costate is to be found. This is done by using *fsolve* to the Eq. (7) and (14). Once they are found the optimal direction unit factor, $\hat{\alpha}$ is calculated. If the solution does not converge another initial guess is provided. After obtaining λ_i and t_f, the initial states, x_0 and λ_0 are propagated, and it is checked if the final result respect the constraints of the problem and if Hamiltonian error is acceptable and through the transversality condition. As asked by the problem, the costate initial guess was $\lambda_{0r,v} \in [-20; 20]$ and the final time $t_f \in [0; 2\pi]$.

$\boldsymbol{\lambda}_{0,r}$	0.497832	−13.818977	0.108153
$\boldsymbol{\lambda}_{0,v}$	5.183553	−10.400418	1.485130
$\lambda_{0,m}$	1.751402		
t_f	2023-10-17-04:05:14.608 UTC		
TOF [days]	141.577842		

Table 5: Time-optimal Earth-Venus transfer solution ($T_{\max} = 800$ mN)(@SUN ECLIPJ2000).

$\|\|\mathbf{r}_f(t_f) - \mathbf{r}_V(t_f)\|\|$	[km]	0.531194
$\|\|\mathbf{v}_f(t_f) - \mathbf{v}_V(t_f)\|\|$	[m/s]	$3.05509 \cdot 10^{-4}$

Table 6: Final state error with respect to Venus' center ($T_{\max} = 800$ mN).

As in the problem is stated that it is not time-dependant and time-optimal solutions, the relative error deviation of the Hamiltonian at each time instance of the propagation is calculated and the results are of order of 10^{-11}.

Part 4

For obtaining the solution in the case of 500 mN, reducing the thrust from 800 mN, numerical continuation has been used. Thus, the thrust is decreased linearly, starting from the initial guess of λ. From the second iteration, the guess for λ is taken as the result of the previous one. The same process as in **Part 3** is repeated for each iteration. The results are shown in Tables 7 and 8, while the plot of the orbit associated to the thrust of 500 mN is shown in Fig. 9.

As expected, with the same departing time, the spacecraft would need a longer period of time than with respect to the case of 800 mN.

$\boldsymbol{\lambda}_{0,r}$	0.123945 $\quad -1.173690 \quad -0.023270$
$\boldsymbol{\lambda}_{0,v}$	2.017028 $\quad 0.155360 \quad -0.119978$
$\lambda_{0,m}$	0.082050
t_f	`20224-01-01-04:20:47.409 UTC`
TOF [days]	217.588638

Table 7: Time-optimal Earth-Venus transfer solution ($T_{\max} = 500$ mN)(@SUN ECLIPJ2000).

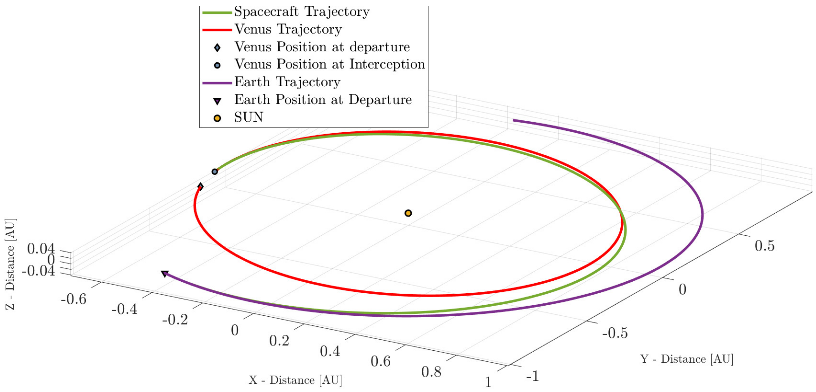


Figure 9: Orbit transfer $T_{max} = 500$mN (@SUN ECLIPJ2000)

$\|\|\mathbf{r}_f(t_f) - \mathbf{r}_V(t_f)\|\|$	[km]	0.22753
$\|\|\mathbf{v}_f(t_f) - \mathbf{v}_V(t_f)\|\|$	[m/s]	$8.303202 \cdot 10^{-5}$

Table 8: Final state error with respect to Venus' center ($T_{\max} = 500$ mN) (@SUN ECLIPJ2000).